

◆ 16.11.8 Control and Feedback Execution (Final — Integrated Version)

Action Operator

$$B : \mathcal{U}_{\{\mathrm{feasible}\}} \rightarrow \mathbb{R}^d$$

satisfying:

$$\|B(u)\| \leq B_{\max}, \quad \|B(u_1) - B(u_2)\| \leq L_B \|u_1 - u_2\|$$

Thus:

B is bounded and Lipschitz continuous.

Closed-Loop Injection

$$F(\Psi_k) \mapsto F(\Psi_k) + B(u_k)$$

Thus:

decisions directly modify system dynamics.

Full Closed-Loop Mapping

$$\Psi_k \rightarrow P_k \rightarrow PAC_k \rightarrow D_k \rightarrow u_k \rightarrow \Psi_{k+1}$$

Thus:

the system operates as a fully closed-loop architecture.

Well-Posed Closed Loop

$$u_k = \pi(\Psi_k) \\ \rightarrow \\ F = F(\Psi_k)$$

Combined with Lipschitz continuity:

$$\exists! \Psi_k$$

Thus:

the closed-loop system admits a unique trajectory.

Stability Preservation

$$|\langle \nabla \mathcal{E}(\Psi_k), B(u_k) \rangle| \leq 0$$

Thus:

$$\dot{\mathcal{L}} \leq -c\|\Psi_k\|^2 + C\sigma^2$$

control actions do not destabilize the system.

Input-to-State Stability (ISS)

$$\begin{aligned} \|\Psi_k\| \\ \leq \\ \beta(\|\Psi_0\|, k) \\ + \gamma(\|u\|) \end{aligned}$$

with bounded input:

$$\begin{aligned} \|u\| &\leq u_{\max} \\ &\rightarrow \\ \|\Psi_k\| &\leq C \end{aligned}$$

Noise Robustness

$$\|\mathbb{E}\|\Psi_k\|^2 \leq C + O(\sigma^2)$$

Thus:

the system remains stable under stochastic disturbances.

State Constraints (Saturation)

$$\Psi \in [\Psi_{\min}, \Psi_{\max}]$$

Thus:

physical constraints are enforced.

Fail-Safe Mechanism

$$\begin{aligned} \|\Psi_k\| &> \Psi_{\mathrm{crit}} \\ &\rightarrow \\ \Psi_k &\rightarrow \Psi_{\mathrm{safe}} \end{aligned}$$

Thus:

catastrophic divergence is prevented.

Real-Time Guarantee

$$T_{\{\mathrm{control}\}} = O(1)$$

Thus:

control execution is constant-time.

Computational Closure

- bounded operations
- finite steps
- no recursive dependence

Thus:

the system is fully implementable.

Compatibility with System Components

- integrates with dynamics (Section 16.11.2)

- preserves Lyapunov stability (Section 16.11.3)
- executes decisions (Section 16.11.7)

Thus:

the control layer completes the closed-loop system.

Implication

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\boxed{
\text{Control is bounded, stable, and guarantees
safe closed-loop execution.}
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SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
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